MLFCS

Matrices: Multiplication & Inversion

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8th December 2022

Lecture attendance code:

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(start recording)

Today's plan

- Definition of matrix multiplication
- Identity for square matrices
- Computing inverse of a matrix

What is a matrix?

For us, matrix is a two-dimensional array whose entries come from a field (say $\mathbb{Q}$ or $\mathbb{R}$)

- We can read it row-wise
- or column-wise

$$\begin{array}{c|ccc} & & & \\ \text{Row 1} & & & \\ \text{Row 2} & & & \end{array} \begin{pmatrix} 1 & 4 & 9 \\ 3 & 5 & 12 \end{pmatrix} = A$$

$$\begin{array}{ccc} C1 & C2 & C3 \end{array}$$

Size of a matrix

- Number of rows
- Number of columns

$$\underset{2\times 3}{A} = (a_{ij}) \quad 1 \le i \le 2$$
$$1 \le j \le 3$$

How to denote entries of a matrix?

a_{ij} = entry in i^{th} row & j^{th} column

We now define matrix multiplication:

- Let A be an $m \times n$ matrix whose entry in row i and column j is given by a_{ij}
- Let B be an $n \times p$ matrix whose entry in row j and column k is given by b_{jk}
- Then the result of multiplying A and B is a $(n \times p)$ matrix C whose entry c_{ik} in row i and column k is given by

$$c_{ik} \quad = \quad \sum_{j=1}^{n} a_{ij} \times b_{jk} \quad = \quad (a_{i1} \times b_{1k}) + (a_{i2} \times b_{2k}) + (a_{i3} \times b_{3k}) + \ldots + (a_{in} \times b_{nk})$$

- That is, the entry in row i and column k of the matrix AB is defined to be the inner product of row i of A with column k of B

$$\underset{A}{\begin{array}{c c} \text{Row 1} & \\ \text{Row 2} & \end{array}\begin{pmatrix} 1 & 3 & 7 \\ 2 & 4 & 9 \end{pmatrix}_{2\times 3}} \times \underset{B}{\begin{array}{c} \begin{pmatrix} 6 & 3 & 4 \\ 7 & 2 & 5 \\ 1 & 3 & 0 \end{pmatrix}_{3\times 3} \\ \begin{array}{ccc} C1 & C2 & C_3 \end{array} \end{array}} = \begin{pmatrix} R_1 \times C_1 & R_1 \times C_2 & R_1 \times C_3 \\ R_2 \times C_1 & R_2 \times C_2 & R_3 \times C_3 \end{pmatrix}$$

Properties of matrix multiplication

Matrix multiplication is not commutative

- For two matrices A and B, it is possible to have $AB \neq BA$

$A_{2\times4}$ $\quad B_{4\times3}$ $\to$ AB is well-defined

$\to$ BA is not

Matrix multiplication is associative

- Let A be a $(m \times n)$ matrix
- Let B be a $(n \times p)$ matrix
- Let C be a $(p \times s)$ matrix
- Then $A(BC) = (AB)C$

Identity matrix for (2×2) matrices

Let $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

Let A be any (2×2) matrix over the field of rational numbers

- Show that $AI = A = IA$

$AI = A$

$IA = A$

Identity matrix for (3×3) matrices

Let $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$

Let A be any (3×3) matrix over the field of rational numbers

- **Show that $AI = A = IA$**

$AI = A$

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

$IA = A$

$\boxed{\text{Exercise}}$

Inverse of a (2×2) matrix

Let A be a (2×2) matrix.

Then a matrix B is said to be inverse of A if $BA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$

How can we find inverse of a specific (2×2) matrix, say

$\boxed{A = \begin{pmatrix} 1 & 2 \\ 3 & 8 \end{pmatrix}?}$

Suppose $B = \begin{pmatrix} r & s \\ t & u \end{pmatrix}$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = BA = \underline{\begin{pmatrix} r & s \\ t & u \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 3 & 8 \end{pmatrix}} = \begin{pmatrix} r+3s & 2r+8s \\ t+3u & 2t+8u \end{pmatrix}$$

First row gives $1 = r+3s$ and $0 = 2r+8s$ $\leftarrow$ 2 equations & 2 variables

Second row gives $0 = t+3u$ and $1 = 2t+8u$ $\swarrow$

Inverse of a matrix need not always exist!

Show that the (2×2) matrix $A = \begin{pmatrix} 1 & 2 \\ 4 & 8 \end{pmatrix}$ does not have an inverse.

Need to show that there is no (2×2) matrix B with entries which are rational numbers such that $BA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Suppose there is $B = \begin{pmatrix} r & s \\ t & u \end{pmatrix}$ s.t $BA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = BA = \begin{pmatrix} r & s \\ t & u \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 4 & 8 \end{pmatrix} = \begin{pmatrix} r+4s & 2r+8s \\ t+4u & 2t+8u \end{pmatrix}$$

$$1 = r+4s$$

$$0 = 2r+8s$$

Left and right inverse are the same!

Not proved in this module!!

Let A be a (2×2) matrix.

Then a matrix B is said to be the left inverse of A if
$BA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

If B is the left inverse of A, then it is also the right inverse of A,
i.e., $BA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow AB = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

If a matrix has an inverse then left inverse is the same as the right inverse. So it is sufficient to just talk about inverse of a matrix without specifying if it is left inverse of right inverse.

Inverse of a (3×3) matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 1 & 0 & 5 \end{pmatrix} \qquad B = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

Does there exist B s.t. $BA = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Does there exist C s.t. $AC = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Consider the system of equations:

$$\boxed{\begin{aligned} 2x_1 - x_2 &= 0 \\ x_1 + x_2 &= 6 \end{aligned}}$$

$$\underbrace{\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}}$$

$$\Downarrow$$

$$2x_1 - x_2 = 0$$
$$x_1 + x_2 = 6$$

$$A\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}, \quad A = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$$

$$A\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}$$

$$A^{-1}A\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = A^{-1}\begin{pmatrix} 0 \\ 6 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 2 & -1 & 0 \\ 1 & 1 & 6 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 1 & 0 & \star \\ 0 & 1 & \heartsuit \end{array}\right)$$

Operations allowed were:

- Rearrange rows
- Multiply a row by any rational number
- Add/subtract any row from another

Consider the system of equations:

$$\begin{pmatrix} 3 & 1 & 5 \\ -3 & 1 & -2 \\ 3 & -1 & 7 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -3 \\ -5 \\ 10 \end{pmatrix}$$

$\underbrace{\qquad}_{A}$ Find A^{-1}

$$3x_1 + x_2 + 5x_3 = -3$$

$$-3x_1 + x_2 - 2x_3 = -5$$

$$3x_1 - x_2 + 7x_3 = 10$$

$$A\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -3 \\ -5 \\ 10 \end{pmatrix}$$

A^{-1} to get $\underbrace{A^{-1}A}_{I}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = A^{-1}\begin{pmatrix} -3 \\ -5 \\ 10 \end{pmatrix}$

$\parallel$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = A^{-1}\begin{pmatrix} -3 \\ -5 \\ 10 \end{pmatrix}$$

Operations allowed are:

- Rearrange rows
- Multiply a row by any rational number
- Add/subtract any row from another

Summary of the lecture

- Definition of matrix multiplication
  - Not always well-defined!
  - Is not commutative in general
  - Is always associative
- Identity for square matrices $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
- Inverse of a matrix
  - Need not always exist...
  - If it exists, then left inverse and right inverse are the same!
- Computing inverse of a matrix
  - Can be computed using Gaussian elimination